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openframeworks – Xcode setup guide:

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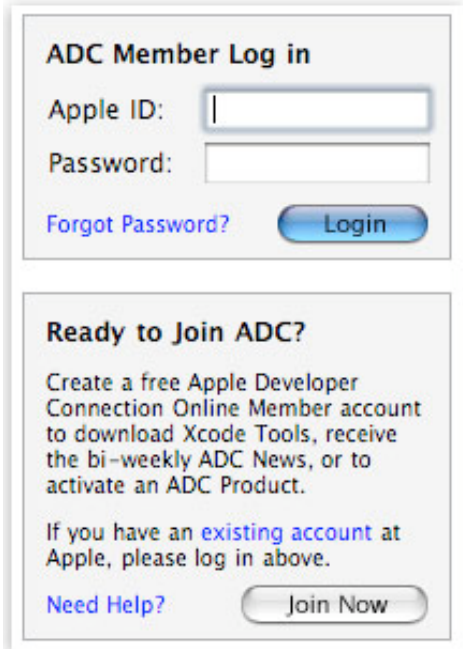
[Notes: Location of your app](#)

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Step 1: Download and install Xcode 2.4.1 or higher

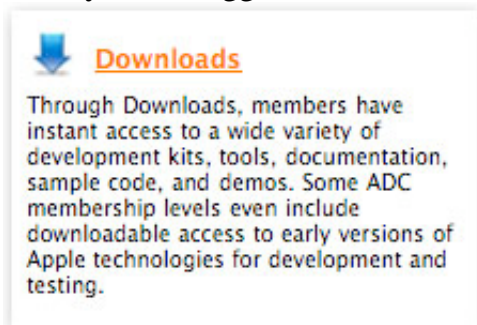
Xcode is Apple's free gcc based IDE for OS X. You can download it from the [Apple Developer Center](#), though you have to be registered as an apple developer. Registration is free and relatively painless.



The screenshot shows two sections of the Apple Developer Connection website. The top section, titled "ADC Member Log in", contains input fields for "Apple ID:" and "Password:", a "Forgot Password?" link, and a blue "Login" button. The bottom section, titled "Ready to Join ADC?", contains text about creating a free account to download Xcode Tools, receive ADC News, or activate an ADC Product. It also includes a link for existing accounts to log in and a "Join Now" button.

login to apple developer connection

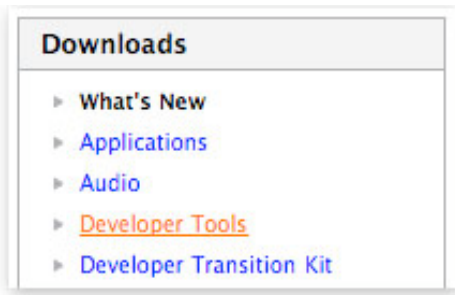
Once you are logged in, click on the link to the download section.



The screenshot shows the "Downloads" section of the Apple Developer Connection website. It features a blue download icon, the word "Downloads" in orange, and text explaining that members have instant access to a wide variety of development kits, tools, documentation, sample code, and demos. It also mentions that some membership levels include downloadable access to early versions of Apple technologies for development and testing.

link to download section

Select the link in the downloads for Developer Tools



link to developer tools

Download the latest version of Xcode

Download Name	File Size	Date Posted
Xcode 2.4.1 (Disk Image)	923.2 MB	31 Oct 2006
Xcode 2.4.1 ReadMe (PDF)	119 KB	31 Oct 2006

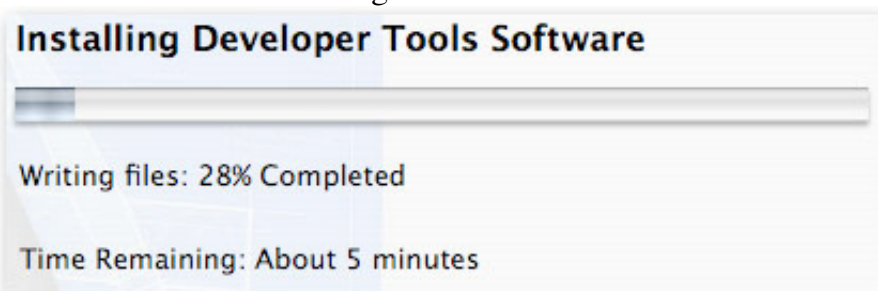
download link for xcode

Once the 900MB download completes, mount the installer and run the XcodeTools package. Click through the installer till you see the customize option. Click it. This gives you the option to skip certain parts of the install. If you are short on disk space you'll probably want to skip the 1.3GB of developer documentation.

Package Name	Action	Size
☒ Developer Tools Software	Upgrade	10.0MB
☒ gcc 4.0	Upgrade	5.0MB
☒ gcc 3.3	Upgrade	0 bytes
☐ Software Development Kits		14.9MB
☐ Java 1.4 Developer Tools		0 bytes
☐ Developer Documentation	Skip	1.3GB
☒ Developer Example Software	Upgrade	646KB
☐ WebObjects		210MB
☐ Cross Development		89.8MB
☐ Automator Shell Script Actions Update	Skip	1.5MB
☐ CHUD Tools	Skip	73.6MB

customize the install

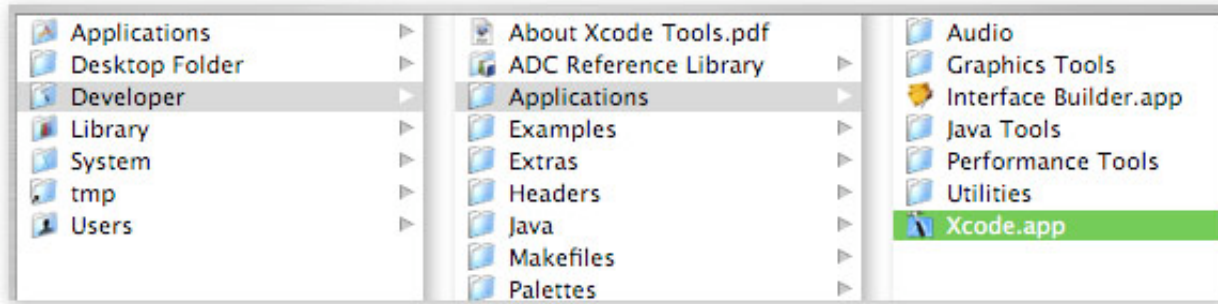
The installer starts installing.



installing

The Developer Tools installer will have installed a folder called *Developer* in the root (highest level) of

your harddrive. The Xcode application is located at */Developer/Applications/Xcode.app*.

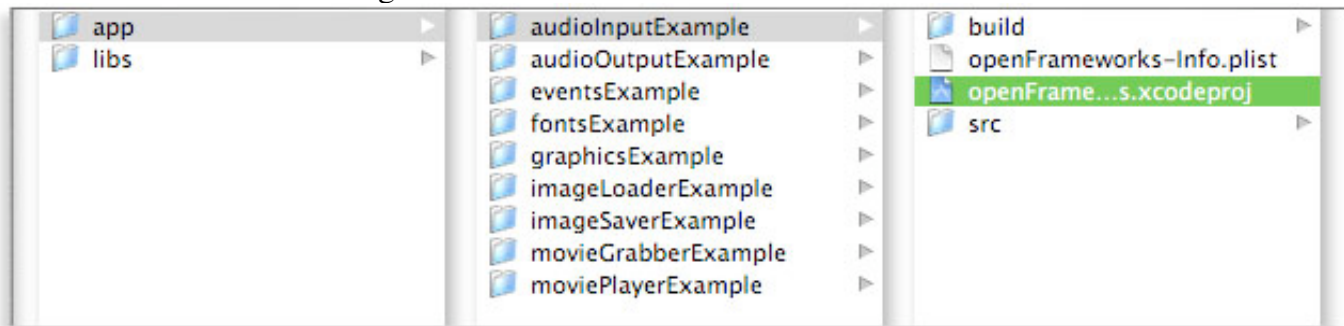


Xcode location

Now that Xcode has been successfully installed on your computer it is time to check out the openFrameworks examples.

Step 2: Using openFrameworks

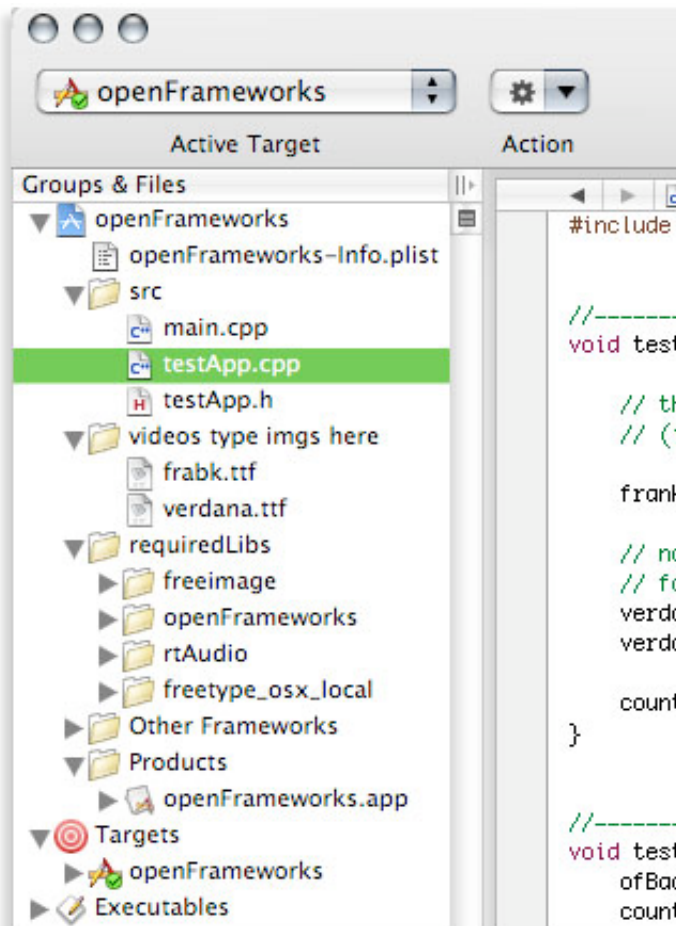
Download the openFrameworks examples from the [download section](#) of this website. Once downloaded and unzipped place the folder in an appropriate location on your harddrive. Unzipped the contents of the folder should look something like this.



examples folder structure

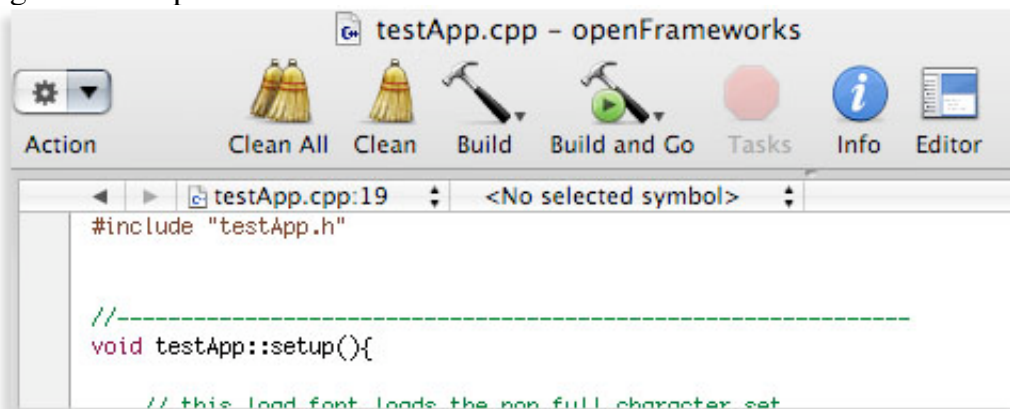
The examples are located within the "app" folder. Within each example folder is the Xcode project file "openFrameworks.xcodeproj". This file will open the Xcode project for that example.

When the example opens in Xcode you will see a list of files in the left column and the content of the currently selected file (in this case "testApp.cpp") in the right panel.

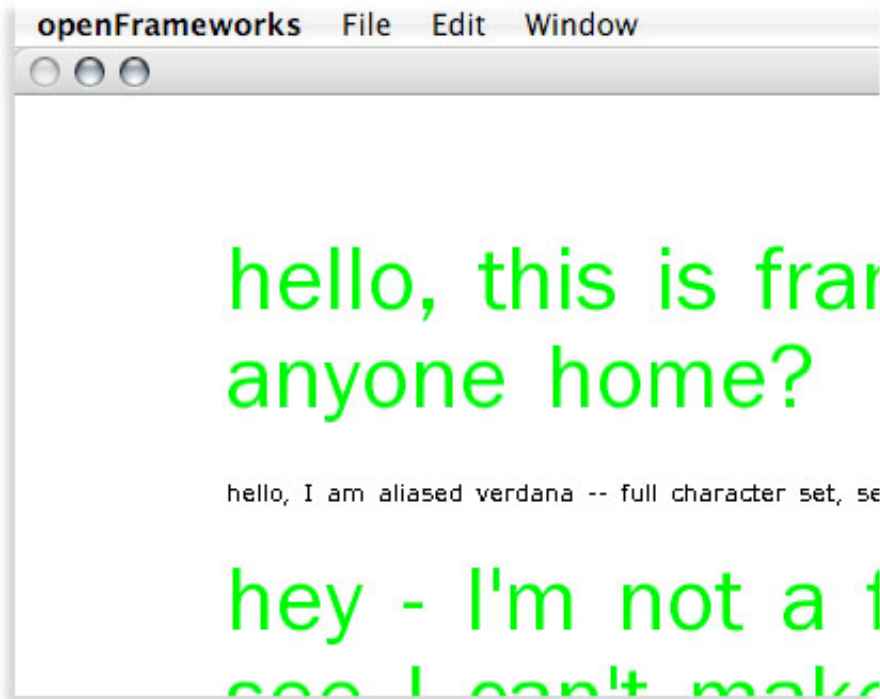


openFrameworks Xcode example project

The first thing you will probably want to do is to compile and run the app to make sure it works. Along the top of the Xcode window you should see a button called *"Build and Go"* this button will compile the example app and as long as there are no errors it should then launch the app successfully. If you get a ton of errors on the first compile - try going to the Build menu and selecting "Clean All Targets" - then try again to compile.



Click *"Build and Go"* To launch the app



The ofType example running

Once the app is running - you can terminate it by hitting the "Esc" key on your keyboard.

Notes: Warnings and Errors

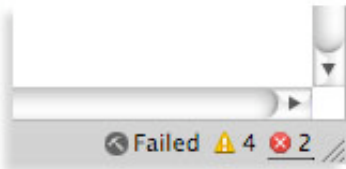
While the app is compiling you'll notice a little circular **progress meter** on the bottom right corner of the Xcode window. This indicates the percentage of the files compiled.

The yellow triangle and the number next to it indicates the number of **warning messages** from the compiler. Most of the time you can ignore the warning messages but if you would like to read them click on the triangle and Xcode will open a separate window which will show them as a list.



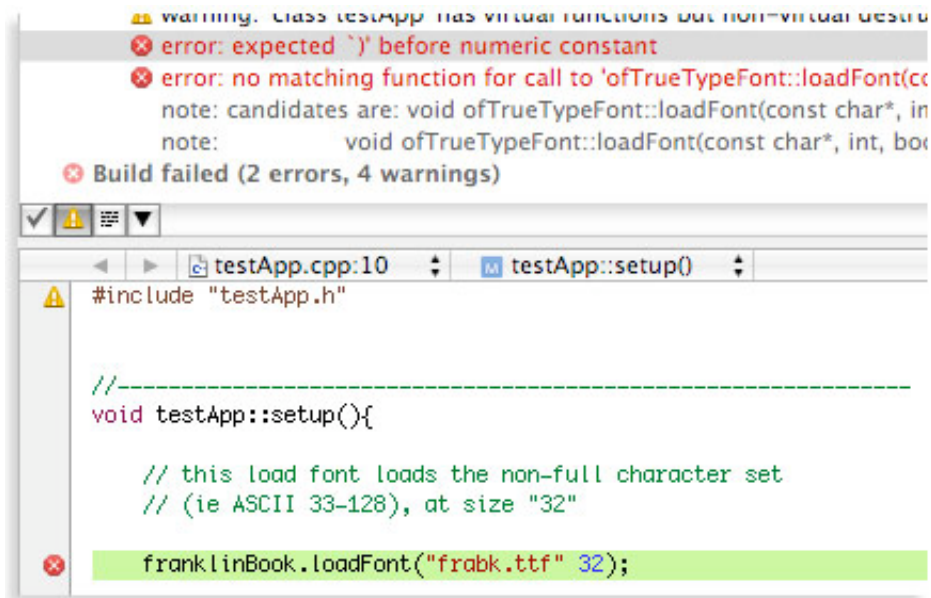
compile progress and warning messages

If the compiler finds **errors** in your code or is just generally unhappy about something the compile will fail and you will see something like this in the bottom right hand corner of your Xcode window.



compile failed due to 2 errors

As with the warning messages, to **view the errors** click on the red circle and Xcode will bring up a new window where it will list the errors and attempt to **highlight the line in the code** where the errors occurred.

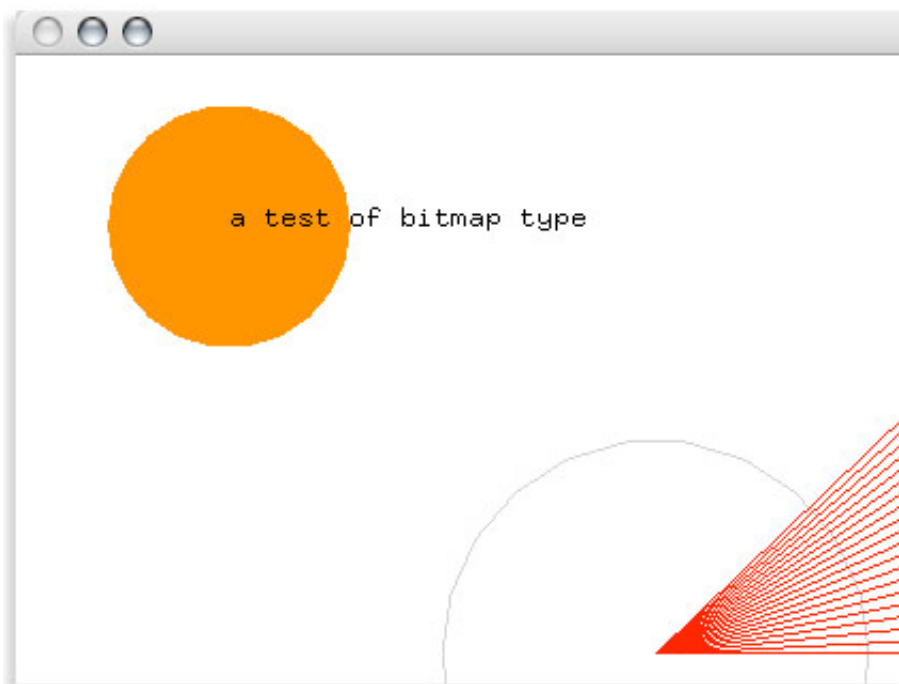


errors highlighted in the code

Bear in mind that the **error message might not always make sense to you** but most of the time just showing you where the error is will be enough for you to realise your mistake.

In the code above, by looking at the highlighted line it is quite easy to see that I forgot to put a coma between "frabk.ttf" (the font name I wanted to load) and 32 (the size I wanted to font to be).

If I put the coma back in both errors go away and the app compiles and launches nicely.



graphics example app launches

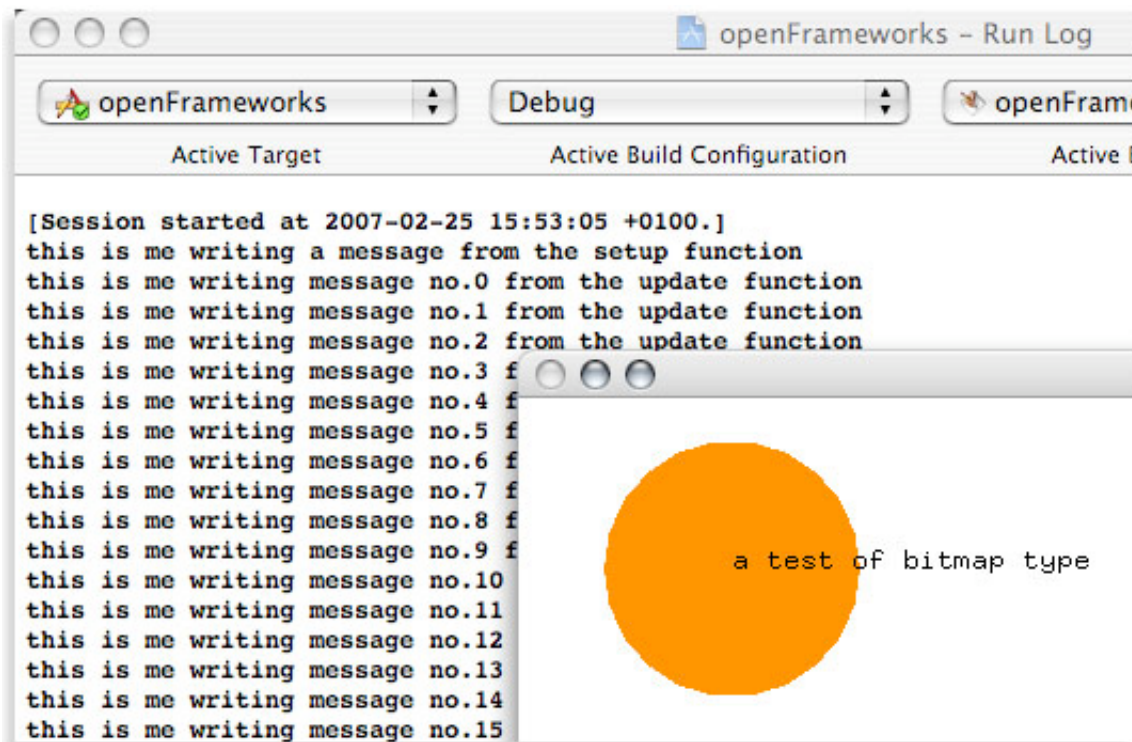
Notes: Run Log

When the Xcode project launches the app it has another window running behind it called the "Run Log". This is a console that will give you **feedback about your app** while it is running.

When the app launches the first thing that is printed to the Run Log is
[Session started at 2007-02-25 15:34:07 +0100.]

When the app exits it will print something like this
openFrameworks has exited with status 0.

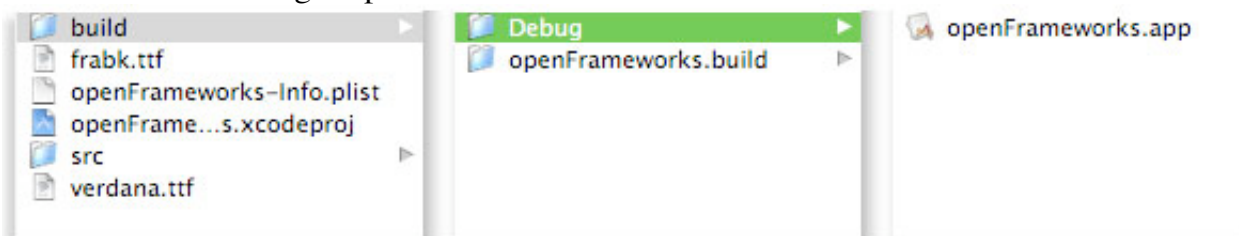
The useful thing about the Run Log is that you can **print** out to it **while your app is running** with the standard C command `"printf"`. Also if there are openFrameworks related messages or errors they will be printed here as well.



the run log for the graphics example

Notes: Location of your app

The application that you are compiling and running with Xcode is called *openFrameworks.app* and it is located in the `"build/Debug/"` folder of each project directory. You can always **run the application directly** by double clicking it. Everything will be the same as when you launch it with Xcode except you won't see the Run Log output.



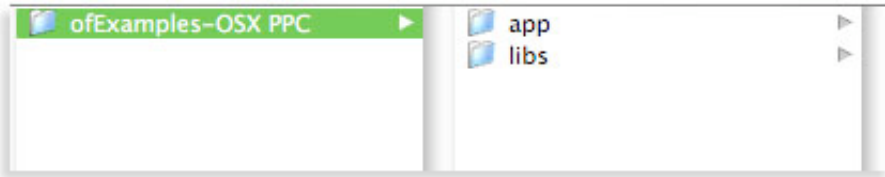
location of the openFrameworks app

Xcode - Bugs and Quirks

Spaces in main project folder name:

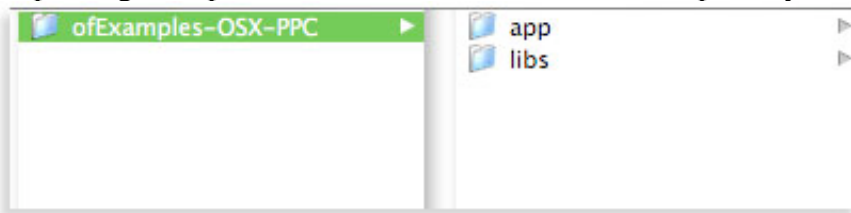
For some reason Xcode freaks out if your main project folder (the one that contains the "*app*" and "*libs*" folder has spaces in the name.

It will give **linking errors** that it can't find certain libraries in the libs folder.



bad folder naming

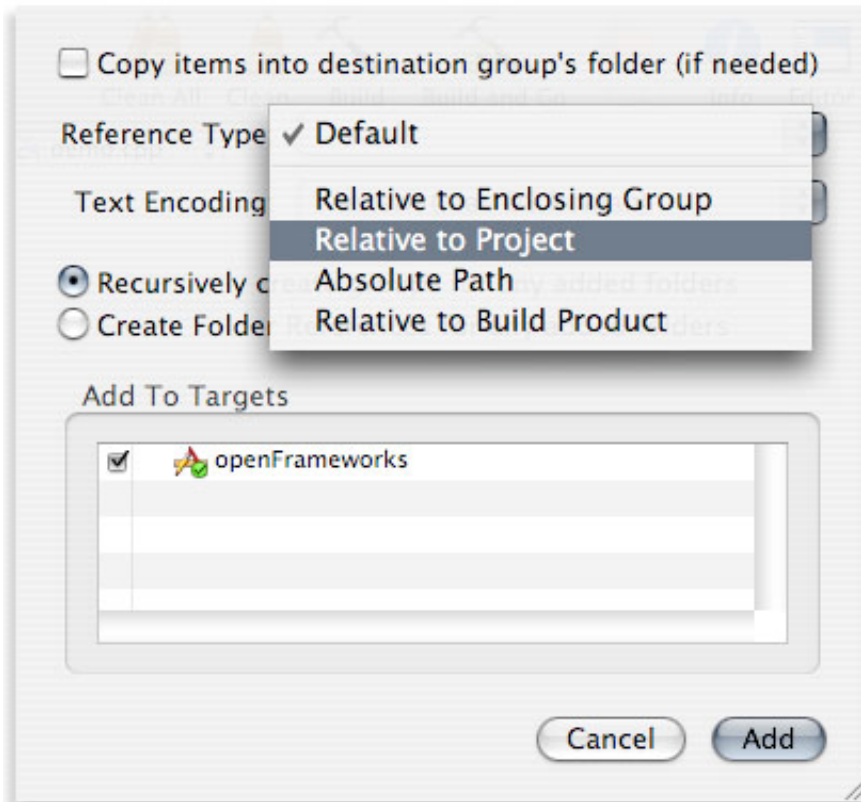
The folder name above, with the space between *OSX* and *PPC* will give you all sorts of problems. Replacing the space with a dash, as below will keep everything happy.



good folder naming

Adding files to project: relative vs absolute path

When you add source code files or libraries to your project Xcode will ask you whether you wish the path to the file to be a relative path or an absolute path.



choose relative to project for all files in your openFrameworks folder

As a rule for **all files** in your main **openFrameworks folder** and its sub-folders make sure you **choose** "relative to project".

If you select **absolute path** then everytime you move your openFrameworks folder around or onto another computer **Xcode will freak out** because your files will have been listed with paths that don't exist anymore.

For example a path like: `"/User/yourname/Documents/openFrameworksProjects"` means that the Xcode project will **only work** if the openFrameworks folder is in that specific place. Obviously this is **not a good thing**.

The **default path** it will give you is **usually an absolute one** so make sure you **change it** to "relative to project" before you click add.

The **only files** that should have **absolute paths** are the **System Frameworks** which you will most likely not need to add.